Comparing HALEU Demand Among Advanced Reactor Fuel Cycle Transitions

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INTRODUCTION

The Light Water Reactors (LWRs) that are currently employed for commercial power in the United States use uranium oxide (UOX) fuel at enrichment levels less than 5%. New reactor designs, such as the Ultra Safe Nuclear Corporation (USNC) Micro Modular Reactor (MMR), will use Low-Enriched Uranium (LEU) fuel enriched between 5-20%, often referred to as High-Assay Low-Enriched Uranium (HALEU). Deploying commercial reactors fueled by a different enrichment level will require changes in multiple front-end resources, such as enrichment feed material and Separative Work Unit (SWU) capacity, to produce the same mass of fuel.

To meet the HALEU fuel demand for advanced reactors, the U.S. Department of Energy (DOE) has proposed two methods to achieve fuel at the required enrichment level: recovery and downblending of High-Enriched Uranium (HEU) fuel from EBR-II and other DOE HEU stockpiles or enriching natural uranium to the desired level [1]. Both of these methods to produce HALEU fuel have limitations. Domestic downblending capabilities and the existing physical supply of HEU both ultimately limit the potential for generating HALEU by downblending. Meanwhile, available SWU capacity and throughput of domestic centrifuge facilities together limit the potential to generate HALEU by enriching natural uranium.

This work simulates multiple fuel cycle scenarios to investigate material demands for the transition to HALEU fueled advanced reactors. Each simulated scenario will investigate the demand of enriched uranium by the reactors and the resources required to meet this demand.

METHODOLOGY

Each fuel cycle scenario is simulated using Cyclus, an agent-based fuel cycle simulator [2]. The agent-based architecture of Cyclus allows for the simulation to treat each facility discretely, including their deployment and decommissioning [2]. The Cyclus Additional Modules Repository (the Cycamore library) provides the archetypes for each of the facility agents deployed in the simulation.

The first scenario only simulates the current fleet of U.S. reactors. This simulation begins in January 1965 and lasts through December 2090 (125 years in total) with one month timesteps. The International Atomic Energy Agency (IAEA) Power Reactor Information System (PRIS) database [3] 2020 Year-end Reactor Status Report provided information about each reactor. This database provides reactor type, rated power level, start-up, and shutdown dates for each of the commercial reactors in the U.S. reactor fleet (as of December 2020). The simulations assume the LWRs operate for 60 years after their initial start date if they have not already been shut down. Only reactors with a power level above 400 MWe were used in the simulation to avoid including prototype and research reactors present in the database. Generic reactor core masses were obtained from [4] and [5].

As Figure 1 shows, other fuel cycle facilities deployed include: a uranium mine (to represent the Cameco Smith Ranch-Highland Mine), a uranium mill (to represent White Mesa Uranium Mill), a conversion facility (to represent the Honeywell Uranium Conversion Facility), an enrichment plant (to represent the Portsmouth Gaseous Diffusion Plant), a fuel fabrication facility (to represent the Westinghouse Fuel Fabrication Facility), and various waste and spent fuel storage and disposal facilities (to represent Yucca Mountain). While there are other fuel cycle facilities in the actual U.S. fuel cycle, this modeling provides a concise total of the resources required by each non-reactor step in the nuclear fuel cycle and simplifies the facilities that are not expected to be deployed or decommissioned often during the specified time frame. Most of these facilities do not have any limits on their production throughputs, except the fuel fabrication plant, which has a front end storage limit of 1 million kg of enriched uranium.

There are multiple reactor agents in each of the scenarios, one agent for each reactor, which are condensed to one facility in Figure 1. The facilities shown in red are used only in the transition scenarios to provide a separate material stream.

Other fuel cycle scenarios simulated include the transition to a future with substantial HALEU reactor deployment under two assumptions about growth in nuclear energy demand: one simulation assuming no growth and the other assuming 1% annual growth. The different scenarios in this work are summarized in Table I. The transition to the new reactor type starts in 2025. These scenarios used the same non-reactor facilities as in the simulation of the current U.S. fuel cycle. HALEU fuel reactors considered in this work include the USNC MMR ™ [6] and the X-Energy Xe-100 ™ Reactor [7][8]. Both of these reactors are designed to use HALEU fuel in the form of Tristructural Isotropic (TRISO) fuel particles. Table II summarizes the key design characteristics of these two reactors.

TABLE I: Fuel cycle scenarios

Scenario No.	Advanced Reactor	Demand Growth
1	None	N/A
2	USNC MMR	No growth
3	X-energy Xe-100	No growth
4	USNC MMR	1% growth
5	X-energy Xe-100	1% growth

We selected these two reactors because they have a high likelihood of being deployed and because published information is available about their designs. Comparing the transitional HALEU demands of these reactors will also inform the

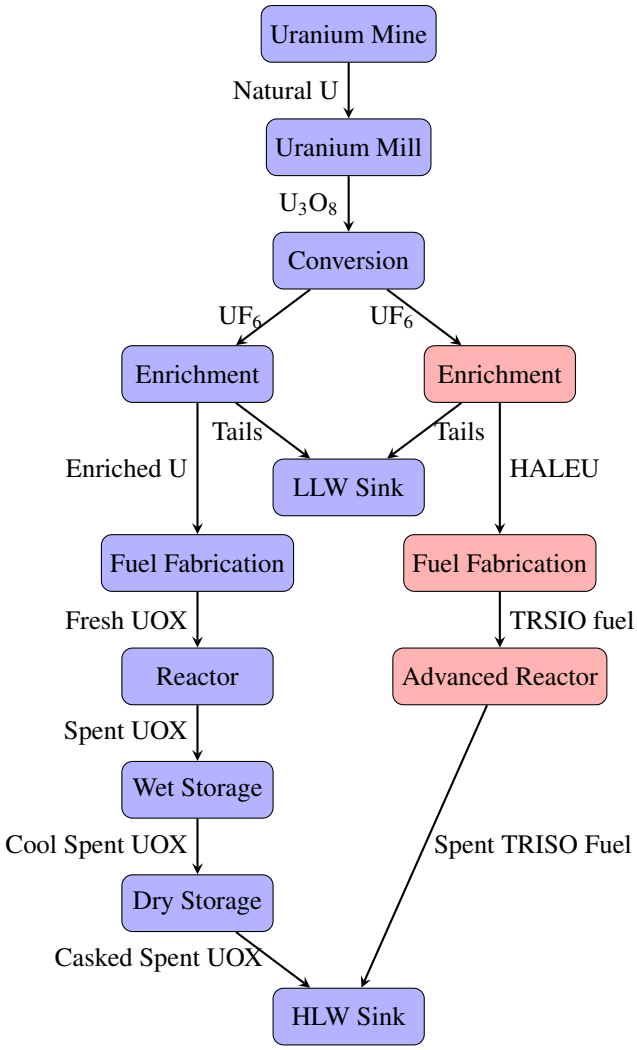

Fig. 1: Fuel cycle facilities and material flow between facilities. Facilities in red are deployed in the transition scenarios.

fuel cycle support implications of deploying reactors with long cycle times versus those utilizing online refueling.

Each of the transition scenarios have been simulated using a single type of advanced reactor, resulting in four additional fuel cycle scenarios and five scenarios in total. Each of the scenarios will be analyzed using Cymetric [9] to determine the resource requirements of the scenario.

RESULTS

The results include the time-dependent resources required for each of the scenarios described. This includes the number of reactors, mass of enriched uranium, and the SWU capacity needed to produce the enriched uranium. Results presented here are for fuel cycle scenarios 1-3.

Figure 2 shows the total number of each type of reactor deployed in each of the simulated scenarios. All scenarios deploy the same number of LWRs. Each of the transitions utilize a Cycamore `GrowthRegion` to control the deployment of the advanced reactor as needed to meet the specified demand,

TABLE II: Mico-reactor design specifications

Design Criteria	USNC MMR™	X-Energy Xe-100™
Reactor type	Modular HTGR	Modular HTGR
Power Output (MWth)	15	200
Enrichment (% ^{235}U)	13	15.5
Cycle Length (years)	20	online refuel
Fuel form	TRISO compacts	TRISO pebbles
Reactor Lifetime	20 years	60 years
Coolant	He	He

while the LWRs are deployed at their specified start date. All of the scenarios have the same deployment schedule for LWRs. To meet the specified power demand of the transition, 5962 MMR reactors and 795 Xe-100 reactors are required.

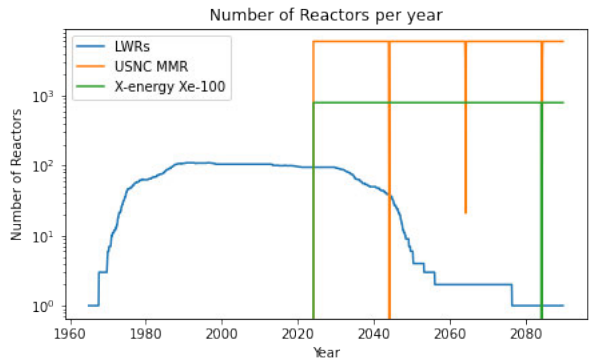

Fig. 2: Deployment of each type of reactor. LWRs follow the same deployment schedule in each scenario.

Figure 3 shows the total energy output from all of the reactors in each scenario. Each of the scenarios have very similar energy outputs from the LWRs, then show large differences in the energy output of the advanced reactors. The energy output in each of the transition scenarios does not match the flat demand starting at 2025 specified by the Cycamore `GrowthRegion`. This is due to the use of mutual transactions in the dynamic resource exchange (DRE) [10], which requires that each material request filled must be filled in full. Thus, we had to be cap the inventory of feed material (UF_6 commodity) that the enrichment facility could store. Placement of this cap to ensure use of both material streams causes small discrepancies in the energy output of the LWRs. Further work is required to determine the cap on feed material inventory to better meet the intended energy demand of the transitions. Scenario 3 more closely matches the energy demand than scenario 2, likely because of the longer lifetime of the Xe-100 reactor. Scenario 2 never reaches the required power level.

Figure 4 shows the amount of HALEU required by scenarios 2 and 3. Both scenarios show a spike in HALEU demand in 1975, which is an artifact of the DRE and is not due to the reactors' fuel requirements. Looking at the demand of HALEU after 2049, scenario 2 requires 4556-6409 kg more than scenario 3, except for a single time step in 2084 when scenario 3 requires almost 69,000 kg more. This sudden spike in enriched uranium demand is due to the deployment of more

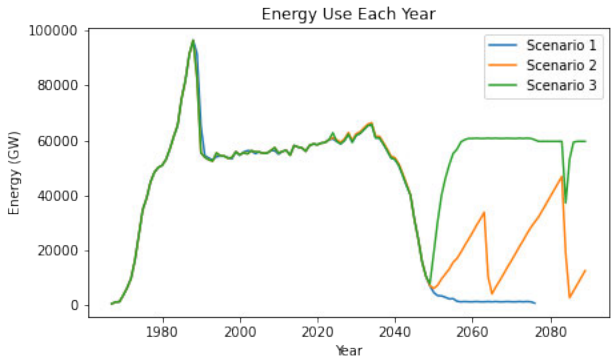

Fig. 3: Total energy output from reactors in each scenario.

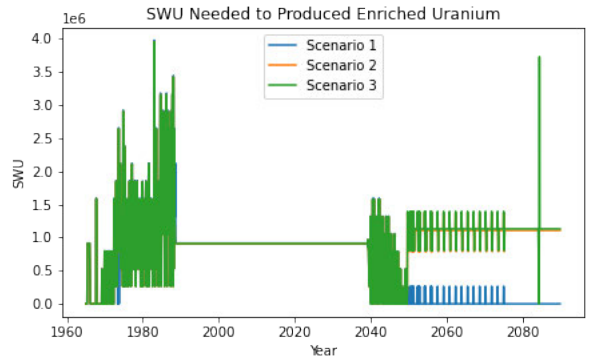

Fig. 5: Enriched uranium demand for reactors.

Xe-100 reactors after the initial set of reactors reach their lifetime.

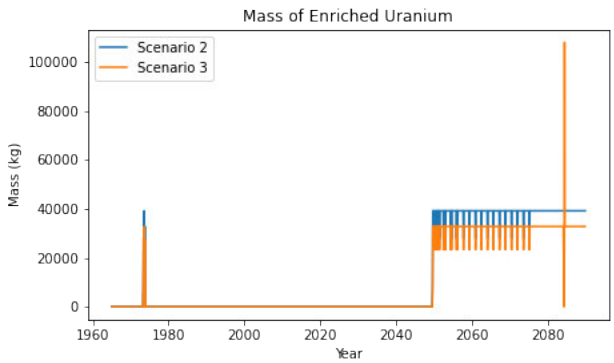

Fig. 4: Enriched uranium demand for reactors.

Finally, Figure 5 shows the SWU required to produce enriched uranium in each scenario. The calculated SWU for scenrios 2 and 3 includes SWU to produce HALEU for the advanced reactor and the LEU to produce fuel for the LWRs. SWU requirements clearly increase from scenario 1 to the transition scenarios once the advanced reactors are fueled. For most of the time steps in which the advanced reactors require fuel, scenario 3 requires 22,881 kg-SWU more than scenario 2, despite scenario 2 requiring a larger mass of HALEU. This is because the Xe-100 reactors in scenario 3 require a higher level of enrichment.

ONGOING WORK

Further work needs to be done to complete the analysis of these fuel cycle scenarios. We will further investigate and adjust characteristics of the DRE to ensure that required power levels are met and simulate the transition scenarios assuming 1% growth in energy demand.

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